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1. Introduction

Photonix Studio is a fictional consumer electronics brand that designs digital cameras and content creation accessories, outsourcing all manufacturing to independent suppliers and selling directly to consumers through its own online store. With no manufacturing capabilities of its own, product quality is determined entirely by supplier selection decisions.

This report presents the design, implementation, and analysis of a relational database built to support informed supplier decision making at Photonix Studio. The database is designed through an entity relationship diagram, implemented in SQLite, populated with synthetic data, and analyzed through SQL to generate insights that address identified challenges and support business objectives.

2. Business Context

2.1 Main Business Activities

Photonix Studio operates as a brand and online retailer, contracting independent suppliers to manufacture products across seven categories: cameras, lenses, lighting, tripods, audio equipment, stabilizers, and accessories, with retail prices ranging from £14 to £599. Five strategically selected suppliers across China, Germany, Vietnam, the United States, and Brazil represent three distinct performance tiers: cheap, average, and premium. Fulfilment is managed through eight warehouses across the UK, USA, Japan, Australia, Germany, Canada, South Korea, and France.

2.2 Core Business Problem

Supplier selection at Photonix Studio has historically been based on two criteria alone: quoted unit manufacturing cost and estimated lead time. Customer feedback and quality outcomes have never been systematically connected to specific supplier and product combinations, leaving the business with no visibility over how these decisions affect product quality or customer satisfaction. Consequently, low-cost suppliers have at times delivered products with high defect rates and elevated return rates, where the financial advantage of lower unit costs is outweighed by the longer-term consequences of returns, reputational damage, and lost repeat purchases.

Christopher (2016) identifies this as a recurring challenge in supply chain management, noting that cost-focused decisions frequently overlook downstream quality costs. Monczka et al. (2015) similarly argue that effective supplier selection must evaluate quality performance alongside cost and delivery to avoid such outcomes.

2.3 Business Objectives and Key Questions

The database provides Photonix Studio with a unified view of supplier performance across cost, lead time, and quality. By integrating product, supplier, order, review, and return data within a single system, the business can identify which suppliers are driving quality failures, quantify their financial consequences, and make selection decisions that reflect the true total cost of supply. The analysis addresses five key questions:

1. Which supplier offers the highest unit margin per product?
2. Which supplier delivers the strongest quality performance per product, based on customer ratings and return rates?
3. How do suppliers compare holistically across cost, lead time, and quality in a single scorecard?
4. Which suppliers are underperforming, and what is the total financial cost of their returns?
5. What net profit does each supplier generate once refunds are deducted from gross margin?

Answering these questions requires relational joins across Supplier, Order_Item, Review, and Return, analytical capability that flat-file alternatives cannot provide

3. Database Design

3.1 Key Entity and Relationship Design

The database is structured around the core operational activities of Photonix Studio, with full entity descriptions in Appendix A.1 and relationship rules in Appendix A.2. The most significant design decisions are discussed below.

Supplier_Product is the central entity. Since a product can be manufactured by multiple suppliers and a supplier can produce multiple products, a many-to-many relationship exists between Supplier and Product. As this cannot be implemented directly without redundancy (Elmasri and Navathe, 2016), Supplier_Product resolves it as an intermediate entity storing the combination-specific attributes of unit cost, minimum order quantity, and lead time, which are needed to evaluate supplier performance. It is assigned a surrogate primary key rather than a composite key, as Order_Item and Inventory reference it directly, simplifying integrity enforcement (Date, 2003).

Sales_Order and Order_Item are held separately because combining them would violate first normal form by repeating item-level attributes across rows. Order-level attributes such as discount rate sit in Sales_Order, while item-level attributes like quantity and selling price sit in Order_Item.

Review and Return are linked to Order_Item rather than Sales_Order, as feedback and returns relate to specific purchased items, enabling analysis at the supplier and product level. Both are optional, expressed as cardinality 1 to 0..1. Payment and Delivery each hold a one-to-one relationship with Sales_Order, while Inventory connects Warehouse and Supplier_Product to track stock per location. The schema observes Third Normal Form throughout. In Order_Item, SellingPrice depends solely on OrderItemID, not on SupplierProductID, eliminating the transitive dependency that would arise from storing price at order level.

3.2 Assumptions in Design

To ensure a clear database structure, key simplifying assumptions have been made in the design. Each order belongs to one customer, and each item belongs to one order. Reviews and returns relate to individual items with at most one each. Every order generates exactly one payment and one delivery record. Refunds equal the full line total after discount. To balance realism and simplicity, auxiliary entities such as inventory are included. Full assumptions are in Appendix A.2.

3.3 Entity-Relationship Diagram

The ERD was first developed in Chen notation at the conceptual level before being translated into Crow's Foot notation for the logical design stage, as Crow's Foot maps more directly to relational schema conventions and communicates cardinality more clearly in a database context (Teorey et al., 2011). Figure 1 presents the Crow's Foot ERD, with the Chen notation version in Appendix A.3.

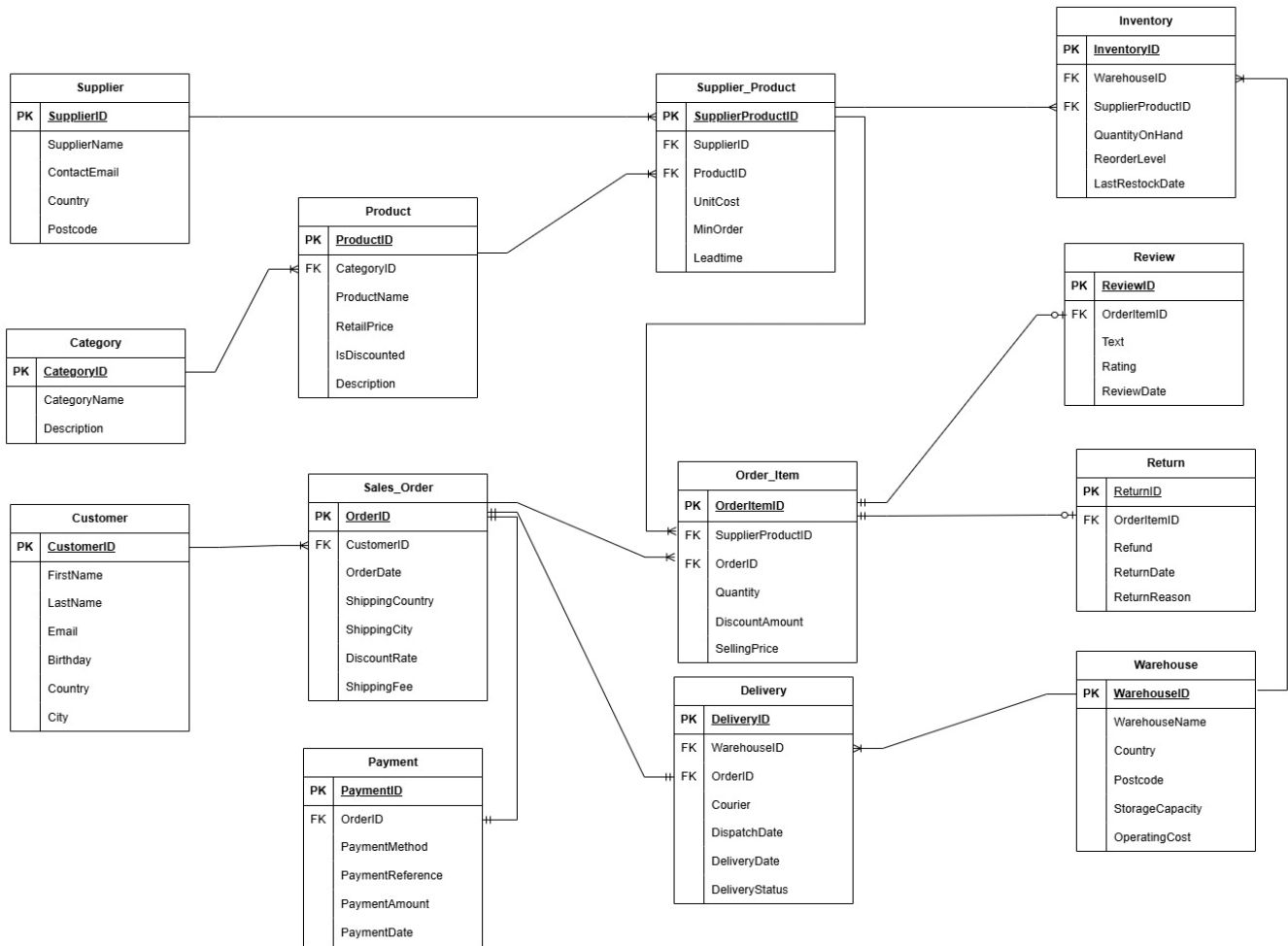


Figure 1. Crow's Foot Entity-Relationship Diagram for the Photonix Studio Database

4. Database Implementation

The database was implemented in SQLite, selected for its simplicity and suitability for this scale of project (SQLite Documentation, 2024). Full schema definitions are in Appendix B.

INTEGER was used for all identifiers, quantities, and the IsDiscounted flag. REAL was applied to all monetary values requiring decimal precision. TEXT was used for names, descriptions, categorical fields, and all date values stored in ISO 8601 format, consistent with SQLite's recommended approach for date handling (SQLite Documentation, 2024). These three types cover all 13 tables without exception.

Foreign key constraints are enforced via PRAGMA foreign_keys = ON, and all surrogate primary keys are declared as INTEGER PRIMARY KEY for efficient indexed lookups. The schema follows Third Normal Form throughout, with all non-key attributes depending on their respective primary keys (Codd,

1972). A CHECK constraint on Review restricts Rating to integers between 1 and 5, preventing malformed data from distorting supplier quality scores.

The database schema was defined using SQL Data Definition Language statements written within the Python script and executed against the SQLite database using `conn.executescript()` via Python's built-in `sqlite3` library. Synthetic data was subsequently loaded by reading from the Python-generated Excel file using `pandas.read_excel()` and inserted into each table using `df.to_sql()`, with referential integrity maintained across all 13 tables. No schema changes were required during implementation. Business insight queries were written and executed separately in the DB Browser SQL editor, with results verified before generating the insights in Section 6.

5. Synthetic Data Generation

Synthetic data was generated using a Python script built with the Faker library, seeded at 42 for reproducibility, with Claude (Anthropic) assisting in scripting and ensuring consistency across all 13 tables. The database contains 12,607 records across 13 tables, as summarized in Table 1.

Table	Record Count
Category	7
Product	33
Supplier	5
Warehouse	8
Supplier Product	78
Customer	800
Inventory	624
Sales Order	2,000
Order Item	3,532
Payment	2,000
Delivery	2,000
Review	836
Return	684

Table 1. Record counts per table

The reference and lookup tables contain fewer than 500 records each by design. All main transactional entities meet or exceed the 500-record threshold.

Five suppliers were structured across three tiers. Shenzhen FastTech (China) and Sul Inovacoes (Brazil) form the cheap tier, with unit costs at 60 to 65% of retail, lead times of 3 to 7 days, return rates of approximately 28%, and ratings averaging 2.0. Reliable Assembly Co (Vietnam) and AmeriTech

Components (USA) form the average tier, with unit costs at 65 to 70% of retail, lead times of 8 to 14 days, return rates between 14 and 19%, and ratings averaging 3.1. Premium Optics Mfg (Germany) forms the premium tier, with unit costs at 70 to 75% of retail, lead times of 14 to 25 days, a return rate of 6%, and ratings averaging 4.5. This structure mirrors real electronics supply chains where manufacturers in China and Brazil compete on price while European manufacturers compete on quality, consistent with GoPro's actual supplier network.

Discounts of 10% for cameras and 8% for other categories apply only to 2024 models, reflecting GoPro's clearance pricing strategy. Shipping fee is free for UK orders above £50, £4.99 below, and £9.99 flat internationally. All payment amounts were verified programmatically against item totals plus shipping fees, ensuring internal consistency. The overall return rate of 19.4% aligns with the consumer electronics category average of 15 to 20% (Euromonitor International, 2022). The review rate of 23.7% was set above the typical e-commerce average to ensure sufficient data for quality analysis (Feefo, 2023). However, the dataset does not capture seasonal variation or full global logistics complexity, limitations that would need addressing before live deployment.

6. Business Insights

SQL queries were developed to analyse the dataset, as shown in Appendix C. The exploratory analysis establishes the financial scale of the business and the core tension the subsequent queries investigate (Figure 2&3). Total revenue from 3,532 items sold is £641,111, with £131,557 refunded. Cameras account for 45% of total revenue yet carry the lowest average rating of 2.66. Tripods achieved the highest rating at 3.47 but contributed the least revenue at £37,370. This inverse relationship between revenue concentration and customer satisfaction points directly to supplier quality as the central business challenge.

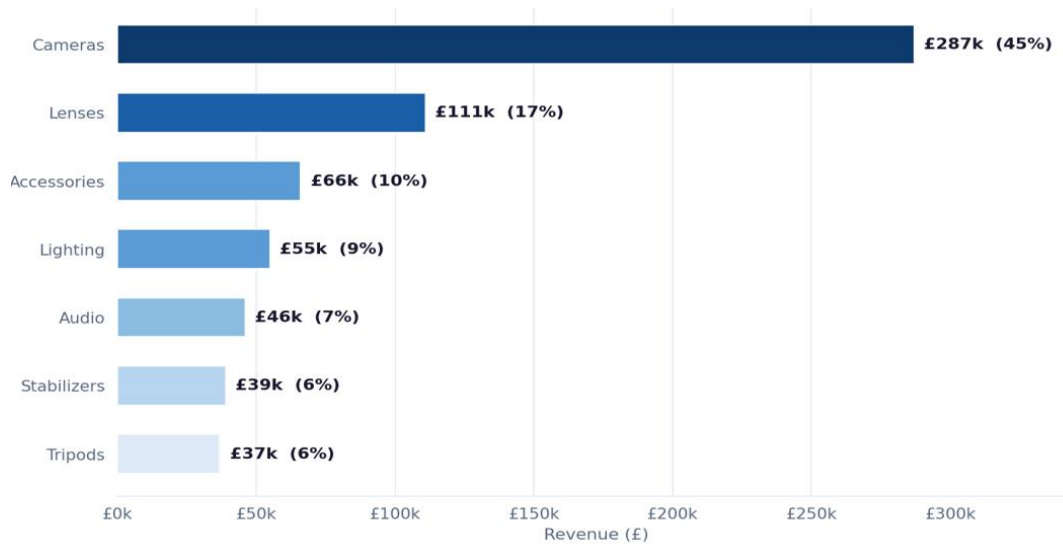


Figure 2- Revenue by Category

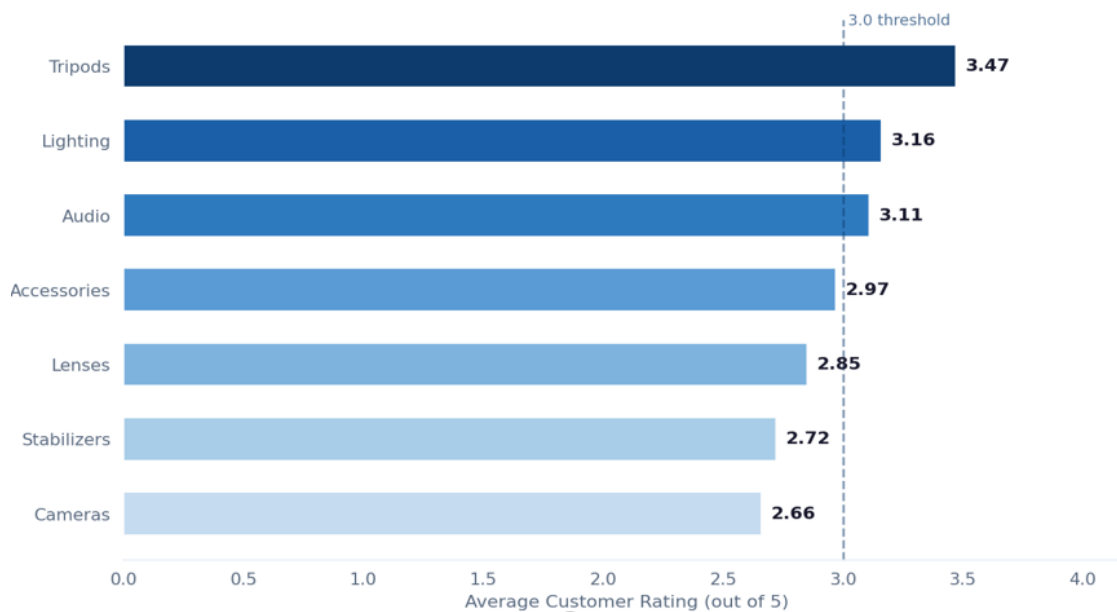


Figure 3- Average Rating by category

Query 1 examines unit margin per product. Shenzhen FastTech and Sul Inovacoes consistently deliver the highest margins, reaching 39.9% on individual products, while Premium Optics Mfg offers the lowest at between 28% and 36%. On cost alone, cheap-tier suppliers appear most financially attractive, which explains their historical preference in selection decisions (Figure 4). However, margin alone fails to capture downstream quality outcomes.

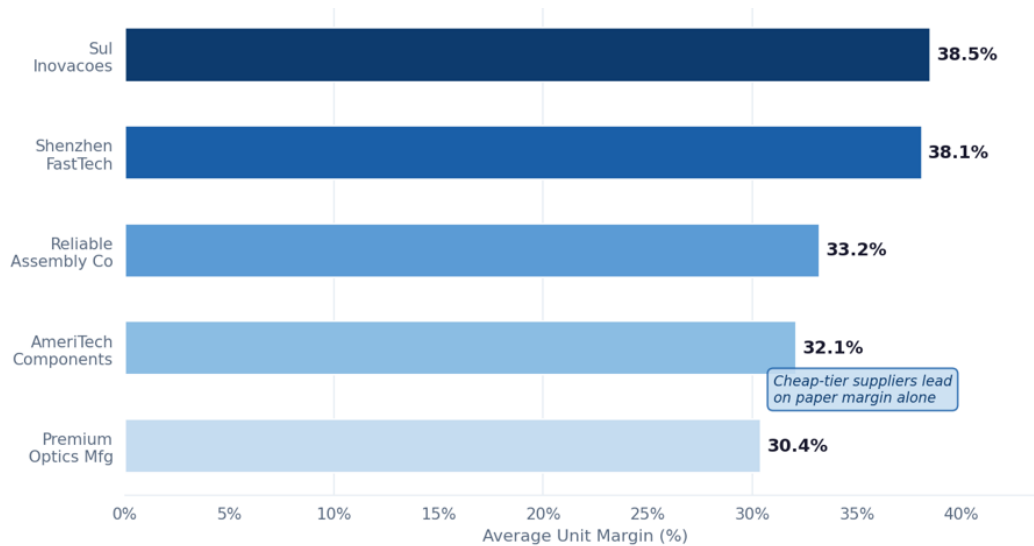


Figure 4 – Average unit Margin by supplier

Query 2 introduces quality by examining average ratings and return rates at the product-supplier level. Premium Optics Mfg dominates the upper end, with the 2025 Photonix 360 Max achieving a 4.00 rating and a 2.6% return rate. In contrast, Sul Inovacoets scores 1.14 on the same product with a 33.3% return rate, and Shenzhen FastTech records a 37.2% return rate on the Photonix Shotgun Microphone S1. Quality failures are concentrated on specific supplier-product combinations, enabling targeted intervention decisions.

Query 3 consolidates cost, lead time, and quality into a full supplier scorecard, visualised across customer ratings (Figure 5) and return rates (Figure 6). Premium Optics Mfg leads on quality with a 4.47 rating and 6.0% return rate but carries the slowest lead time at 20 days. Shenzhen FastTech and Sul Inovacoets are the fastest at approximately 5 days but show poor quality, with ratings near 2.0 and return rates above 28%. AmeriTech Components represents the strongest compromise, achieving the highest average unit margin at £71.61 alongside a moderate 14.4% return rate.

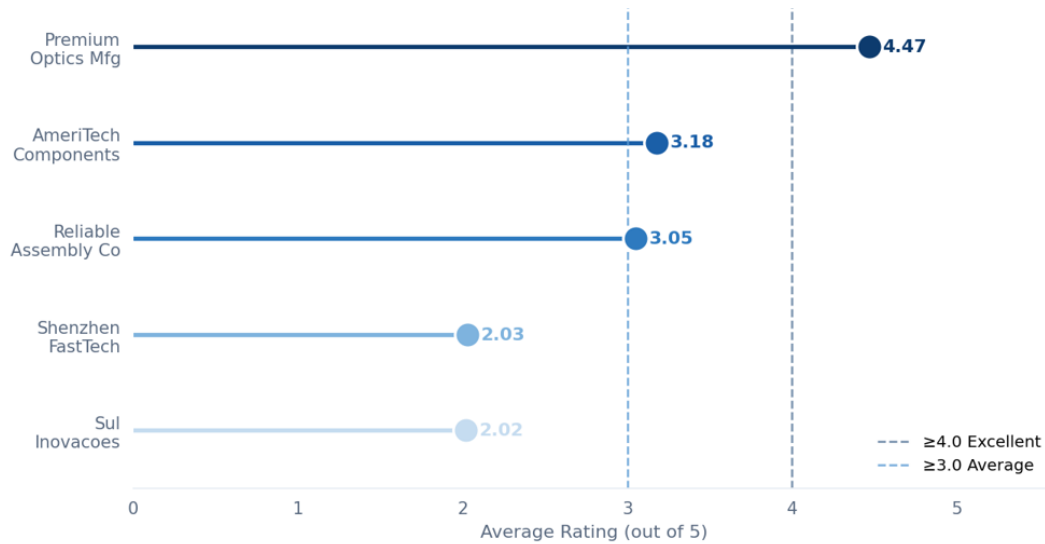


Figure 5 — Customer Rating

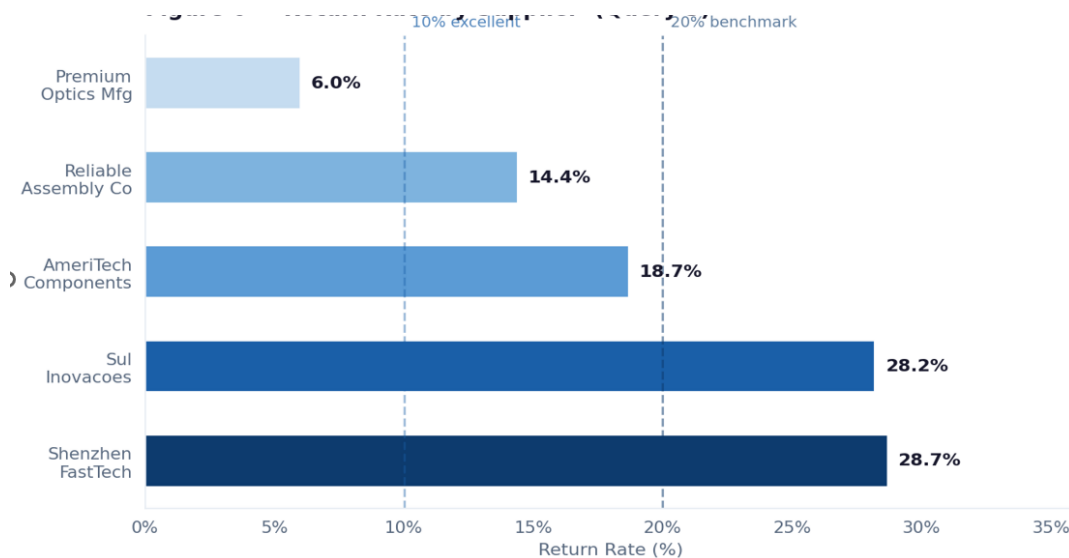


Figure 6 — Return Rate

Query 4 formalises three tiers of supplier classification by applying rating and return rate thresholds (Figure 7). Premium Optics Mfg is the only supplier classified as Excellent, meeting the threshold of a rating above 4.0 and a return rate below 10%. Reliable Assembly Co and AmeriTech Components are classified as Average. Shenzhen FastTech and Sul Inovacoos are both Underperforming, together generating £86,110 in refunds and accounting for 65% of the total refunds issued. This indicates that underperforming suppliers are the primary source of financial loss and should be prioritised in supplier management decisions.

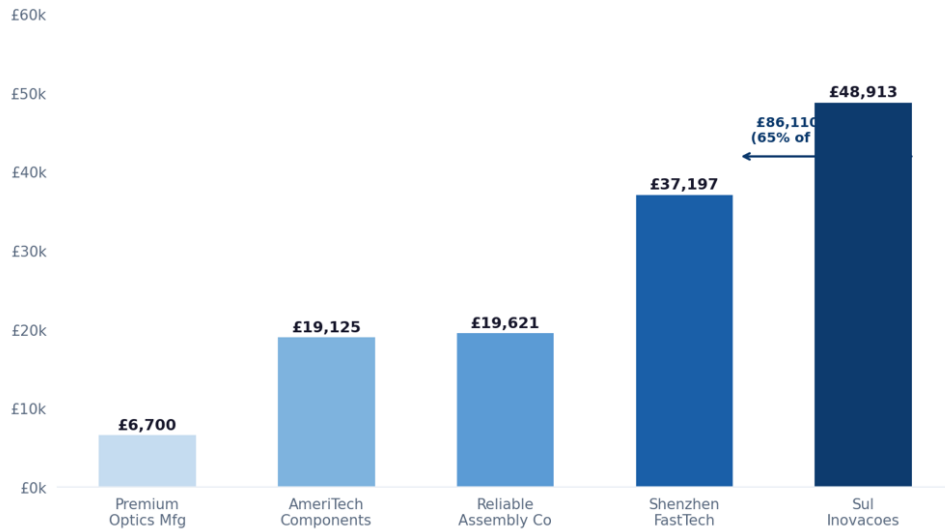


Figure 7 — Total Refunds by Supplier

Query 5 completes the analysis by calculating net profit per supplier after refunds are deducted (Figure 8). Sul Inovacoos generates the highest revenue at £166,130 but ranks third in net profit at £11,499, as refunds eliminate most of its gross margin. Shenzhen FastTech ranks last at £6,974 in net profit despite generating £129,538 in revenue. In contrast, Premium Optics Mfg generates the highest net profit at £21,482, more than three times that of Shenzhen FastTech, despite lower revenue. Once quality costs are accounted for, the apparent cost advantage of cheap-tier suppliers is reversed entirely, confirming that current selection criteria understate the true cost of supply.

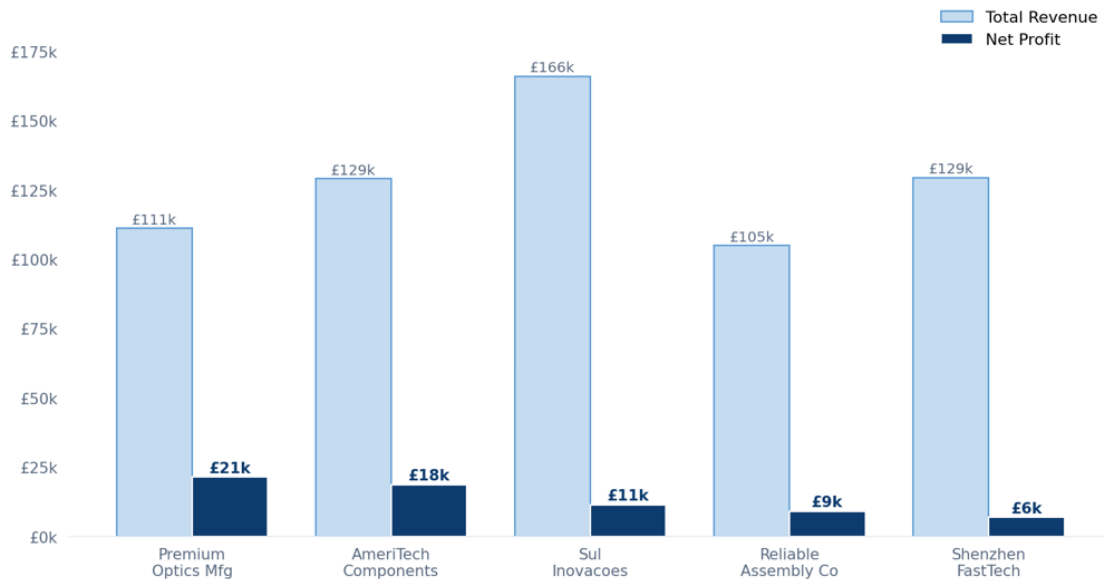


Figure 8 — Revenue vs Net Profit

The five queries form a deliberate sequence: Query 1 establishes the apparent case for cheap-tier q22suppliers; Queries 2 and 3 introduce quality evidence that challenges it; Query 4 quantifies the financial cost; and Query 5 completes the inversion. The supplier ranked cheapest generates the lowest net profit once quality costs are measured.

7. Conclusion

This report has demonstrated that integrating supplier, product, order, review, and return data within a single relational database provides Photonix Studio with the supplier visibility its current decision-making process lacks. The analysis reveals a consistent and financially material pattern in which cheap-tier suppliers offer the highest unit margins on paper but generate refund costs that eliminate and reverse that advantage. Shenzhen FastTech and Sul Inovacoes together account for 65% of all refunds despite operating at the lowest unit costs. The data support a clear strategic recommendation: reallocating production toward Premium Optics Mfg and AmeriTech Components would improve both net profitability and customer satisfaction. Future development could incorporate live supplier performance tracking and seasonal demand data to further strengthen selection decisions.

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Appendix

Appendix A: Database design

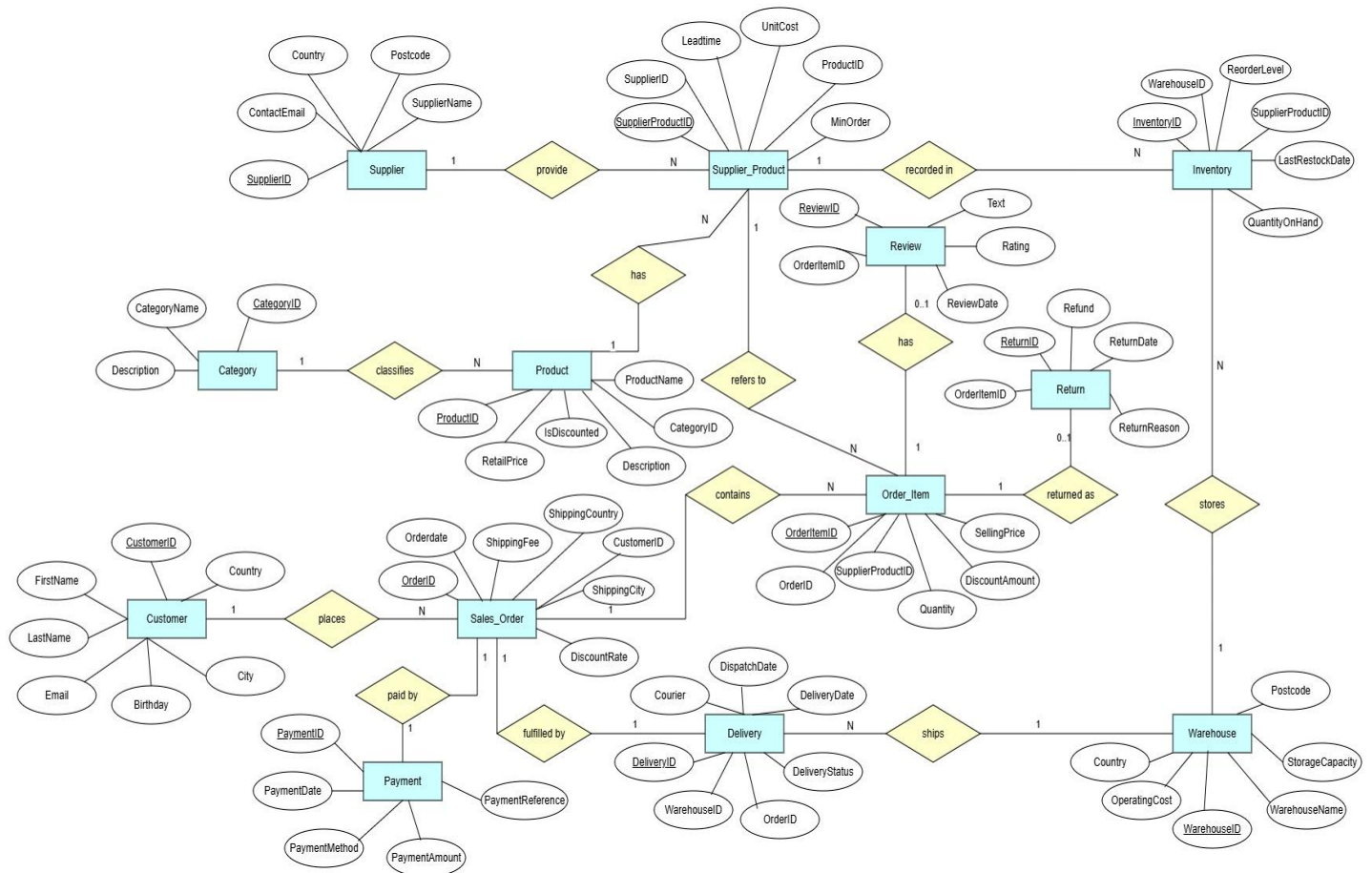
Appendix A.1 Explanation for entities

Entity	Role in the Database
Customer	Stores information about customers who place orders, allowing the business to analyse customer behaviour and purchasing patterns.
Sales_Order	Records each customer order and links transactions to individual customers.
Order_Item	Represents the individual items within each order, enabling item-level sales and performance analysis.
Product	Stores information about the products sold by Photonix Studio.
Category	Groups products into categories to support product organisation and category-level analysis.
Supplier	Stores information about external manufacturers or suppliers.
Supplier_Product	Resolves the many-to-many relationship between suppliers and products, and stores supplier-specific attributes such as cost and lead time.
Review	Captures customer feedback on purchased items, supporting quality and satisfaction analysis.
Return	Records returned items, allowing the business to analyse return patterns and supplier-related quality issues.
Warehouse	Stores information about storage locations used by the business.
Inventory	Tracks the quantity of supplier products held in each warehouse.
Payment	Stores payment information associated with customer orders.
Delivery	<i>Records dispatch and delivery information for each order.</i>

Appendix A.2 Relationship rules

No.	Relationship	Cardinality	Description
1	Category – Product	1 : M	Each product belongs to one category, while each category can contain multiple products.
2	Supplier – Product (via Supplier_Product)	M : N	A supplier can provide multiple products, and a product can be supplied by multiple suppliers. This many-to-many relationship is resolved through the Supplier_Product table. Each record refers to one supplier and one product.
3	Customer – Sales_Order	1 : M	A customer can place multiple orders, but each order belongs to only one customer.
4	Sales_Order – Order_Item	1 : M	A sales order can contain multiple order items, while each order item belongs to one sales order.
5	Supplier_Product – Order_Item	1 : M	Each order item refers to one supplier-product record, while a supplier-product record can appear in multiple order items.
6	Order_Item – Review	1 : 0..1	An order item may have zero or one review, while each review must correspond to exactly one order item.
7	Warehouse – Inventory	1 : M	A warehouse can have multiple inventory records, while each inventory record belongs to one warehouse.
8	Supplier_Product – Inventory	1 : M	A supplier-product record can appear in multiple warehouses through multiple inventory records, while each inventory record refers to one supplier-product record.
9	Warehouse – Delivery	1 : M	Each delivery is dispatched from one warehouse, while a warehouse can handle multiple deliveries.
10	Sales_Order – Delivery	1 : 1	Each order corresponds to exactly one delivery, and each delivery is associated with one order.
11	Sales_Order – Payment	1 : 1	Each order corresponds to exactly one payment, and each payment is associated with one order.
12	Order_Item – Return	1 : 0..1	An order item may be returned at most once, while each return record must correspond to one order item.

Appendix A.3 Chen's version of ERD



Appendix B: SQL Schema

```
CREATE TABLE Category (  
CategoryID INTEGER PRIMARY KEY,  
CategoryName TEXT NOT NULL,  
Description TEXT  
);
```

```
CREATE TABLE Product (  
ProductID INTEGER PRIMARY KEY,  
CategoryID INTEGER NOT NULL,  
ProductName TEXT NOT NULL,  
RetailPrice REAL NOT NULL,  
Description TEXT,  
IsDiscounted INTEGER NOT NULL DEFAULT 0,  
FOREIGN KEY (CategoryID) REFERENCES Category(CategoryID)  
);
```

```
CREATE TABLE Supplier (  
SupplierID INTEGER PRIMARY KEY,  
SupplierName TEXT NOT NULL,  
ContactEmail TEXT,  
Country TEXT,  
Postcode TEXT  
);
```

```
CREATE TABLE Warehouse (  
WarehouseID INTEGER PRIMARY KEY,  
WarehouseName TEXT NOT NULL,  
Country TEXT,  
Postcode TEXT,  
StorageCapacity INTEGER,  
OperatingCost REAL  
);
```

```
CREATE TABLE Supplier_Product (  
SupplierProductID INTEGER PRIMARY KEY,  
SupplierID INTEGER NOT NULL,  
ProductID INTEGER NOT NULL,  
UnitCost REAL NOT NULL,  
MinOrder INTEGER,  
Leadtime INTEGER,  
FOREIGN KEY (SupplierID) REFERENCES Supplier(SupplierID),  
FOREIGN KEY (ProductID) REFERENCES Product(ProductID)  
);
```

```
CREATE TABLE Inventory (  
InventoryID INTEGER PRIMARY KEY,  
WarehouseID INTEGER NOT NULL,  
SupplierProductID INTEGER NOT NULL,  
QuantityOnHand INTEGER NOT NULL DEFAULT 0,  
ReorderLevel INTEGER,  
LastRestockDate TEXT,  
FOREIGN KEY (WarehouseID) REFERENCES Warehouse(WarehouseID),  
FOREIGN KEY (SupplierProductID) REFERENCES Supplier_Product(SupplierProductID)  
);
```

```
CREATE TABLE Customer (  
CustomerID INTEGER PRIMARY KEY,  
FirstName TEXT NOT NULL,  
LastName TEXT NOT NULL,  
Email TEXT UNIQUE,  
Birthday TEXT,  
Country TEXT,  
City TEXT  
);
```

```
CREATE TABLE Sales_Order (  

```

```

OrderID INTEGER PRIMARY KEY,
CustomerID INTEGER NOT NULL,
OrderDate TEXT NOT NULL,
ShippingCountry TEXT,
ShippingCity TEXT,
DiscountRate TEXT NOT NULL DEFAULT '0%',
ShippingFee REAL NOT NULL DEFAULT 0,
FOREIGN KEY (CustomerID) REFERENCES Customer(CustomerID)
);

```

```

CREATE TABLE Order_Item (
OrderItemID INTEGER PRIMARY KEY,
SupplierProductID INTEGER NOT NULL,
OrderID INTEGER NOT NULL,
Quantity INTEGER NOT NULL DEFAULT 1,
SellingPrice REAL NOT NULL,
DiscountAmount REAL NOT NULL DEFAULT 0,
FOREIGN KEY (SupplierProductID) REFERENCES Supplier_Product(SupplierProductID),
FOREIGN KEY (OrderID) REFERENCES Sales_Order(OrderID)
);

```

```

CREATE TABLE Payment (
PaymentID INTEGER PRIMARY KEY,
OrderID INTEGER NOT NULL UNIQUE,
PaymentMethod TEXT,
PaymentReference TEXT,
PaymentAmount REAL NOT NULL,
PaymentDate TEXT,
FOREIGN KEY (OrderID) REFERENCES Sales_Order(OrderID)
);

```

```

CREATE TABLE Delivery (
DeliveryID INTEGER PRIMARY KEY,
WarehouseID INTEGER NOT NULL,

```

```

OrderID INTEGER NOT NULL UNIQUE,
Courier TEXT,
DispatchDate TEXT,
DeliveryDate TEXT,
DeliveryStatus TEXT,
FOREIGN KEY (WarehouseID) REFERENCES Warehouse(WarehouseID),
FOREIGN KEY (OrderID) REFERENCES Sales_Order(OrderID)
);

```

```

CREATE TABLE Review (
ReviewID INTEGER PRIMARY KEY,
OrderItemID INTEGER NOT NULL UNIQUE,
Text TEXT,
Rating INTEGER CHECK (Rating BETWEEN 1 AND 5),
ReviewDate TEXT,
FOREIGN KEY (OrderItemID) REFERENCES Order_Item(OrderItemID)
);

```

```

CREATE TABLE Return (
ReturnID INTEGER PRIMARY KEY,
OrderItemID INTEGER NOT NULL UNIQUE,
Refund REAL,
ReturnDate TEXT,
ReturnReason TEXT,
FOREIGN KEY (OrderItemID) REFERENCES Order_Item(OrderItemID)
);

```

The full synthetic data generation script is included in the submitted ZIP file as Data_Management.ipynb.

Appendix C: SQL Queries

Appendix C.1: Exploratory

Exploratory 1 — Portfolio overview

```
1  -- Exploratory 1 — Portfolio overview
2
3  -- Overview of the entire database
4  -- Confirms all data loaded correctly across key entities
5  SELECT
6      (SELECT COUNT(*) FROM Product) AS TotalProducts,
7      (SELECT COUNT(*) FROM Supplier) AS TotalSuppliers,
8      (SELECT COUNT(*) FROM Customer) AS TotalCustomers,
9      (SELECT COUNT(*) FROM Sales_Order) AS TotalOrders,
10     (SELECT COUNT(*) FROM Order_Item) AS TotalItemsSold,
11     (SELECT COUNT(*) FROM Review) AS TotalReviews,
12     (SELECT COUNT(*) FROM Return) AS TotalReturns;
```

	TotalProducts	TotalSuppliers	TotalCustomers	TotalOrders	TotalItemsSold	TotalReviews	TotalReturns	
1	33	5	800	2000	3532	836	684	

Exploratory 2 — Financial overview

```
12
13  -- Exploratory 2 — Financial overview
14  -- Total revenue shows how much customers paid after discounts
15  -- Total refunds shows the financial cost of poor supplier quality
16  -- Return rate shows what percentage of items sold were returned
17  SELECT
18      ROUND(SUM(oi.SellingPrice * oi.Quantity - oi.DiscountAmount), 2) AS TotalRevenue,
19      ROUND(SUM(CASE WHEN ret.ReturnID IS NOT NULL THEN ret.Refund ELSE 0 END), 2) AS TotalRefunds,
20      ROUND(COUNT(ret.ReturnID) * 100.0 / COUNT(oi.OrderItemID), 1) AS ReturnRatePct
21  FROM Order_Item oi
22  LEFT JOIN Return ret ON oi.OrderItemID = ret.OrderItemID;
```

	TotalRevenue	TotalRefunds	ReturnRatePct	
1	641111.46	131556.64	19.4	

Exploratory 3 — Revenue by category

```

24 -- Exploratory 3 – Revenue by category
25 -- Identifies which categories contribute most to the business
26 -- Average rating shows customer satisfaction at category level
27 SELECT
28     c.CategoryName,
29     COUNT(oi.OrderItemID) AS ItemsSold,
30     ROUND(SUM(oi.SellingPrice * oi.Quantity - oi.DiscountAmount), 2) AS Revenue,
31     ROUND(AVG(r.Rating), 2) AS AvgRating
32 FROM Order_Item oi
33 JOIN Supplier_Product sp ON oi.SupplierProductID = sp.SupplierProductID
34 JOIN Product p ON sp.ProductID = p.ProductID
35 JOIN Category c ON p.CategoryID = c.CategoryID
36 LEFT JOIN Review r ON oi.OrderItemID = r.OrderItemID
37 GROUP BY c.CategoryName
38 ORDER BY Revenue DESC

```

	CategoryName	ItemsSold	Revenue	AvgRating
1	Cameras	876	287087.7	2.66
2	Lenses	429	110890.04	2.85
3	Accessories	968	65822.2	2.97
4	Lighting	399	54966.24	3.16
5	Audio	322	45875.68	3.11
6	Stabilizers	225	39099.6	2.72
7	Tripods	313	37370.0	3.47

Exploratory 4 — Top products by units sold

```

40 -- Exploratory 4 – Top products by units sold
41 -- High volume products have the greatest impact on quality costs
42 -- if a supplier performs poorly on these it hurts the business most
43 SELECT
44     p.ProductName,
45     c.CategoryName,
46     COUNT(oi.OrderItemID) AS UnitsSold,
47     ROUND(SUM(oi.SellingPrice * oi.Quantity - oi.DiscountAmount), 2) AS Revenue
48 FROM Order_Item oi
49 JOIN Supplier_Product sp ON oi.SupplierProductID = sp.SupplierProductID
50 JOIN Product p ON sp.ProductID = p.ProductID
51 JOIN Category c ON p.CategoryID = c.CategoryID
52 GROUP BY p.ProductName
53 ORDER BY UnitsSold DESC
54 LIMIT 10;

```

	ProductName	CategoryName	UnitsSold	Revenue
1	Photonix Prime Lens 35mm f1.8	Lenses	128	15191.04
2	2024 Photonix 360 Max	Cameras	127	47891.7
3	Photonix Fast Charger USB-C	Accessories	125	8612.12
4	Photonix Wireless Lavalier Mic Duo	Audio	122	20618.0
5	2024 Photonix Creator Vlog Camera	Cameras	121	30383.1
6	Photonix SD Card 64GB Extreme	Accessories	120	5014.92
7	2024 Photonix Alpha Flagship	Cameras	120	40932.0

Exploratory 5 — Supplier contribution to portfolio

```

56 -- Exploratory 5 – Supplier contribution to portfolio
57 -- Shows how much revenue and volume each supplier drives
58 -- Sets the stage for the detailed supplier comparison queries
59 SELECT
60     s.SupplierName,
61     s.Country,
62     COUNT(DISTINCT sp.ProductID) AS ProductsSupplied,
63     COUNT(oi.OrderItemID) AS TotalItemsSold,
64     ROUND(SUM(oi.SellingPrice * oi.Quantity - oi.DiscountAmount), 2) AS TotalRevenue
65 FROM Supplier s
66 JOIN Supplier_Product sp ON s.SupplierID = sp.SupplierID
67 LEFT JOIN Order_Item oi ON sp.SupplierProductID = oi.SupplierProductID
68 GROUP BY s.SupplierName
69 ORDER BY TotalRevenue DESC;

```

	SupplierName	Country	ProductsSupplied	TotalItemsSold	TotalRevenue
1	Sul Inovacoes	Brazil	18	870	166129.92
2	Shenzhen FastTech	China	15	658	129538.08
3	AmeriTech Components	USA	13	585	129149.88
4	Premium Optics Mfg	Germany	18	781	111287.88
5	Reliable Assembly Co	Vietnam	14	638	105005.7

Appendix C.2: Business Insights

Query 1 — Which supplier gives the highest unit margin per product?

```

73 -- Query 1 – Which supplier gives the highest unit margin per product?
74 -- Unit margin = retail price minus unit cost
75 -- Higher margin means more profit per unit sold before quality costs
76 SELECT
77     p.ProductName,
78     s.SupplierName,
79     s.Country,
80     p.RetailPrice,
81     sp.UnitCost,
82     ROUND(p.RetailPrice - sp.UnitCost, 2) AS UnitMargin,
83     ROUND((p.RetailPrice - sp.UnitCost) / p.RetailPrice * 100, 1) AS MarginPct
84 FROM Supplier_Product sp
85 JOIN Product p ON sp.ProductID = p.ProductID
86 JOIN Supplier s ON sp.SupplierID = s.SupplierID
87 ORDER BY p.ProductName, UnitMargin DESC;

```

	ProductName	SupplierName	Country	RetailPrice	UnitCost	UnitMargin	MarginPct
1	2024 Photonix 360 Max	Shenzhen FastTech	China	419.0	271.02	147.98	35.3
2	2024 Photonix 360 Max	AmeriTech Components	USA	419.0	280.11	138.89	33.1
3	2024 Photonix 360 Max	Reliable Assembly Co	Vietnam	419.0	286.39	132.61	31.6
4	2024 Photonix Action Camera V-Basic	Shenzhen FastTech	China	179.0	109.59	69.41	38.8
5	2024 Photonix Action Camera V-Basic	Reliable Assembly Co	Vietnam	179.0	117.27	61.73	34.5
6	2024 Photonix Alpha Flagship	Shenzhen FastTech	China	379.0	229.77	149.23	39.4
7	2024 Photonix Alpha Flagship	AmeriTech Components	USA	379.0	256.81	122.19	32.2

```

73 -- Query 1 -- Which supplier gives the highest unit margin per product?
74 -- Unit margin = retail price minus unit cost
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76 SELECT
77     p.ProductName,
78     s.SupplierName,
79     s.Country,
80     p.RetailPrice,
81     sp.UnitCost,
82     ROUND(p.RetailPrice - sp.UnitCost, 2) AS UnitMargin,
83     ROUND((p.RetailPrice - sp.UnitCost) / p.RetailPrice * 100, 1) AS MarginPct
84 FROM Supplier_Product sp
85 JOIN Product p ON sp.ProductID = p.ProductID
86 JOIN Supplier s ON sp.SupplierID = s.SupplierID
87 ORDER BY p.ProductName, UnitMargin DESC;

```

	ProductName	SupplierName	Country	RetailPrice	UnitCost	UnitMargin	MarginPct
8	2024 Photonix Alpha Flagship	Reliable Assembly Co	Vietnam	379.0	258.07	120.93	31.9
9	2024 Photonix Creator Vlog Camera	Sul Inovacoes	Brazil	279.0	170.48	108.52	38.9
10	2024 Photonix Creator Vlog Camera	AmeriTech Components	USA	279.0	192.64	86.36	31.0
11	2025 Photonix 360 Max	Sul Inovacoes	Brazil	529.0	321.72	207.28	39.2
12	2025 Photonix 360 Max	Premium Optics Mfg	Germany	529.0	377.44	151.56	28.7
13	2025 Photonix Action Camera V-Basic	Sul Inovacoes	Brazil	219.0	131.75	87.25	39.8
14	2025 Photonix Action Camera V-Basic	AmeriTech Components	USA	219.0	144.9	74.1	33.8

Query 2 — Which supplier has the best quality per product?

```

89 -- Query 2 -- Which supplier has the best quality per product?
90 -- Combines average rating and return rate per supplier per product
91 -- Only includes combinations with at least 5 sales for reliability
92 SELECT
93     p.ProductName,
94     s.SupplierName,
95     COUNT(oi.OrderItemID) AS TotalSold,
96     ROUND(AVG(r.Rating), 2) AS AvgRating,
97     COUNT(ret.ReturnID) AS TotalReturns,
98     ROUND(COUNT(ret.ReturnID) * 100.0 / COUNT(oi.OrderItemID), 1) AS ReturnRatePct
99 FROM Order_Item oi
100 JOIN Supplier_Product sp ON oi.SupplierProductID = sp.SupplierProductID
101 JOIN Product p ON sp.ProductID = p.ProductID
102 JOIN Supplier s ON sp.SupplierID = s.SupplierID
103 LEFT JOIN Review r ON oi.OrderItemID = r.OrderItemID

```

	ProductName	SupplierName	TotalSold	AvgRating	TotalReturns	ReturnRatePct
1	2024 Photonix 360 Max	Reliable Assembly Co	50	3.64	6	12.0
2	2024 Photonix 360 Max	AmeriTech Components	37	3.09	6	16.2
3	2024 Photonix 360 Max	Shenzhen FastTech	40	2.1	10	25.0
4	2024 Photonix Action Camera V-Basic	Reliable Assembly Co	57	3.16	8	14.0
5	2024 Photonix Action Camera V-Basic	Shenzhen FastTech	52	2.0	11	21.2
6	2024 Photonix Alpha Flagship	Reliable Assembly Co	42	3.33	11	26.2
7	2024 Photonix Alpha Flagship	AmeriTech Components	36	3.3	5	13.9


```

94     s.SupplierName,
95     COUNT(oi.OrderItemID) AS TotalSold,
96     ROUND(AVG(r.Rating), 2) AS AvgRating,
97     COUNT(ret.ReturnID) AS TotalReturns,
98     ROUND(COUNT(ret.ReturnID) * 100.0 / COUNT(oi.OrderItemID), 1) AS ReturnRatePct
99 FROM Order_Item oi
100 JOIN Supplier_Product sp ON oi.SupplierProductID = sp.SupplierProductID
101 JOIN Product p ON sp.ProductID = p.ProductID
102 JOIN Supplier s ON sp.SupplierID = s.SupplierID
103 LEFT JOIN Review r ON oi.OrderItemID = r.OrderItemID
104 LEFT JOIN Return ret ON oi.OrderItemID = ret.OrderItemID
105 GROUP BY p.ProductName, s.SupplierName
106 HAVING TotalSold >= 5
107 ORDER BY p.ProductName, AvgRating DESC;
108

```

	ProductName	SupplierName	TotalSold	AvgRating	TotalReturns	ReturnRatePct	
8	2024 Photonix Alpha Flagship	Shenzhen FastTech	42	2.0	11	26.2	
9	2024 Photonix Creator Vlog Camera	AmeriTech Components	57	3.6	7	12.3	
10	2024 Photonix Creator Vlog Camera	Sul Inovacoes	64	2.36	21	32.8	
11	2025 Photonix 360 Max	Premium Optics Mfg	38	4.0	1	2.6	
12	2025 Photonix 360 Max	Sul Inovacoes	46	1.14	16	33.3	
13	2025 Photonix Action Camera V-Basic	AmeriTech Components	47	2.59	8	17.0	
14	2025 Photonix Action Camera V-Basic	Sul Inovacoes	54	2.08	11	20.4	

Query 3 — Full supplier scorecard across cost, lead time and quality

```

109 -- Query 3 — Full supplier scorecard across cost, lead time and quality
110 -- Holistic view for direct side-by-side supplier comparison
111 SELECT
112     s.SupplierName,
113     s.Country,
114     ROUND(AVG(sp.UnitCost), 2) AS AvgUnitCost,
115     ROUND(AVG(sp.Leadtime), 1) AS AvgLeadtimeDays,
116     ROUND(AVG(r.Rating), 2) AS AvgRating,
117     ROUND(COUNT(ret.ReturnID) * 100.0 / COUNT(oi.OrderItemID), 1) AS ReturnRatePct,
118     ROUND(AVG(p.RetailPrice - sp.UnitCost), 2) AS AvgUnitMargin
119 FROM Supplier s
120 JOIN Supplier_Product sp ON s.SupplierID = sp.SupplierID
121 JOIN Product p ON sp.ProductID = p.ProductID
122 LEFT JOIN Order_Item oi ON sp.SupplierProductID = oi.SupplierProductID
123 LEFT JOIN Review r ON oi.OrderItemID = r.OrderItemID

```

	SupplierName	Country	AvgUnitCost	AvgLeadtimeDays	AvgRating	ReturnRatePct	AvgUnitMargin	
1	Premium Optics Mfg	Germany	91.22	20.0	4.47	6.0	34.6	
2	Reliable Assembly Co	Vietnam	105.26	10.7	3.18	18.7	50.86	
3	AmeriTech Components	USA	146.97	11.5	3.05	14.4	71.61	
4	Shenzhen FastTech	China	115.89	5.3	2.03	28.7	69.12	
5	Sul Inovacoes	Brazil	108.59	5.2	2.02	28.2	66.82	

Query 4 — Supplier tier classification and refund impact

```

128 -- Query 4 — Supplier tier classification and refund impact
129 -- Ranks suppliers as Excellent, Average, or Underperforming
130 -- Shows the total financial cost of returns per supplier
131 SELECT
132     s.SupplierName,
133     s.Country,
134     COUNT(DISTINCT sp.ProductID) AS ProductsCovered,
135     ROUND(AVG(r.Rating), 2) AS AvgRating,
136     ROUND(COUNT(ret.ReturnID) * 100.0 / COUNT(oi.OrderItemID), 1) AS ReturnRatePct,
137     ROUND(SUM(ret.Refund), 2) AS TotalRefundsIssued,
138     CASE
139         WHEN AVG(r.Rating) >= 4.0 AND COUNT(ret.ReturnID) * 100.0 / COUNT(oi.OrderItemID) < 10 THEN 'Excellent'
140         WHEN AVG(r.Rating) >= 3.0 AND COUNT(ret.ReturnID) * 100.0 / COUNT(oi.OrderItemID) < 20 THEN 'Average'
141         ELSE 'Underperforming'
142     END AS SupplierTier

```

	SupplierName	Country	ProductsCovered	AvgRating	ReturnRatePct	TotalRefundsIssued	SupplierTier
1	Premium Optics Mfg	Germany	18	4.47	6.0	6700.2	Excellent
2	Reliable Assembly Co	Vietnam	14	3.18	18.7	19621.42	Average
3	AmeriTech Components	USA	13	3.05	14.4	19124.68	Average
4	Shenzhen FastTech	China	15	2.03	28.7	37197.36	Underperforming
5	Sul Inovacoes	Brazil	18	2.02	28.2	48912.98	Underperforming

Query 5 — Revenue and net profit per supplier after refunds

```

152 -- Query 5 — Revenue and net profit per supplier after refunds
153 -- Profit = revenue minus cost of goods sold
154 -- Net profit = profit minus refunds issued
155 -- Shows which suppliers are actually most valuable after quality costs
156 SELECT
157     s.SupplierName,
158     s.Country,
159     COUNT(oi.OrderItemID) AS TotalItemsSold,
160     ROUND(SUM(oi.SellingPrice * oi.Quantity - oi.DiscountAmount), 2) AS TotalRevenue,
161     ROUND(SUM(sp.UnitCost * oi.Quantity), 2) AS TotalCOGS,
162     ROUND(SUM(oi.SellingPrice * oi.Quantity - oi.DiscountAmount) - SUM(sp.UnitCost * oi.Quantity), 2) AS TotalProfit,
163     ROUND(SUM(ret.Refund), 2) AS TotalRefunds,
164     ROUND(SUM(oi.SellingPrice * oi.Quantity - oi.DiscountAmount) - SUM(sp.UnitCost * oi.Quantity) - SUM(COALESCE(ret.Refund, 0)), 2) AS NetProfit,
165     FROM Order_Item oi

```

	SupplierName	Country	TotalItemsSold	TotalRevenue	TotalCOGS	TotalProfit	TotalRefunds	NetProfit
1	Premium Optics Mfg	Germany	781	111287.88	83106.05	28181.83	6700.2	21481.63
2	AmeriTech Components	USA	585	129149.88	91185.1	37964.78	19124.68	18840.1
3	Sul Inovacoes	Brazil	870	166129.92	105718.08	60411.84	48912.98	11498.86
4	Reliable Assembly Co	Vietnam	638	105005.7	76196.74	28808.96	19621.42	9187.54
5	Shenzhen FastTech	China	658	129538.08	85366.96	44171.12	37197.36	6973.76